MILITARY SPECIFICATION

FIBER, RED (SHEET), FOR SMALL ARMS BUFFER DISKS

This specification is approved for use by Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and grade of hard, red fiber in sheet form suitable for buffer disks in small arms.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL
UU-P-31 - Paper; general Specifications and Methods of Testing

STANDARDS

FEDERAL
Fed. Test Method Std. No. 406 - Plastic, Methods of Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

/ FSC 9420/

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MILITARY
MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM) Standards
ASTM D 638 - Tensile Properties of Plastics
ASTM D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials
ASTM D 3951 - Practice for Commercial Packaging

Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 Material. Sheets shall be composed of chemically treated and hardened pure cellulose fiber, and shall be uniform in color, texture, density, and thickness.

3.2 Chemical composition. The chemical composition shall conform to the requirements shown in table I.

Table I - Chemical composition

Constituent	Requirement
	percent
Water soluble chloride as $ZnCl_2$, maximum	0.2
Ash, maximum	3.5
Cellulose, minimum	96.3

3.3 Physical requirements. Material shall have the physical properties shown in table II when tested in accordance with 4.4.2.

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Table II - Physical requirements

Property	Requirement
Hardness (Rockwell M), minimum	50
Specific gravity	1.30
Tensile strength (lengthwise), minimum, p.s.i.	8500
Tensile strength (crosswise), minimum, p.s.i.	6500
Water absorption (0.125 ± 0.062" thickness), maximum, percent	15
Permanent set (total for 22 disks stacked)	
0.969" diameter, 9000 load lbs., max., inch	0.050
1.094" diameter, 9550 load lbs., max., inch	0.050

3.4 Dimensions. Dimension and tolerances shall be in accordance with the applicable drawing of the part or as specified in the contract (see 6.2).

3.5 Workmanship. All details of workmanship shall be in accordance with the best commercial practice. Finished sheets shall be clean, uniform, have smooth plane surfaces, and be free from defects which may affect their value for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 Lot. A lot shall consist of all material of the same thickness, produced at the same time under the same operating conditions and procedures.

4.2.2 Sampling for examination. A random sample of fiber sheets shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. In terms of defects per 10 untis, the acceptable quality level shall be 2.5 for major defects.

4.2.3 Sampling for tests. Unless otherwise specified (see 6.2) one sheet 12 inches by 15 inches shall be selected from each lot for tests specified in 4.4.2.

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4.3 Examination. Each sample sheet selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table III.

Table III - Classification of defects

Categories	Defects
Major	Dimensions of material
101	not as specified.
102	Workmanship not as specified.

4.4 Tests.

4.4.1 Chemical Tests

4.4.1.1 Zinc chloride. A test specimen consisting of 5 grams of finely divided particles, such as millings or filings, of the fiber shall be placed in a weighed Soxhlet cup. The cup and specimen shall be placed in an oven and dried for 2 hours at 216 - 230°F. and then cooled to room temperature in a desiccator and weighed as quickly as possible to avoid accumulation of moisture. The specimen shall be extracted with distilled water for 3 hours using a Soxhler extractor (see figure 1). After the extraction has been completed, the chloride may be determined either gravimetrically or volumetrically by silver nitrate using any standard method. If the test is completed volumetrically, a blank test shall be made on an equal volume of distilled water and potassium chromate solution and the result shall be subtracted from that of the specimen. The net result shall be calculated in terms of zinc chloride.

4.4.1.2 Ash. The percentage of ash shall be determined in accordance with method 202 of UU-P-31.

4.4.1.3 Cellulose. The cellulose content shall be calculated by subtracting from 100 the sum of the percentages of ash and zinc chloride.

4.4.2 Physical tests.

4.4.2.1. Conditioning of test speciments. Test specimens shall be conditioned by holding for 20 hours at a temperatore of 217°F., plus or minus 5°F., and then cooled to room temperature (75°F. plus or minus 5°F.) in a desicccator prior to conducting the test specified in 4.4.2.2 through 4.4.2.6.

4.4.2.2 Hardness. Any convenient sample having a smooth natural surface may be used. For thicknesses under 0.109 inch two or more pieces, as may be required, are placed one on the other to produce a thickness of at least 0.125 inch. The Rockwell hardness determination shall be made in accordance with ASTM D 785.

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4.4.2.3 Specific gravity. The specimen, which may be of any suitable size and simple geometric shape, shall be weighed in air (weight 1) and then suspended in water and quickly reweighed (weight 2). The specific gravity shall be calculated as follows:

$$\frac{\text{Weight 1}}{\text{Weight 1-Weight 2}} = \text{Specific Gravity}$$

This or any other suitable method may be used to determine the specific gravity.

4.4.2.4 Tensile strength. The tensile strength shall be determined in accordance with ASTM-D 638. Three specimens cut lenghwise and three specimens cut crosswise of the sheet shall be tested. The average of these three results in each case shall be taken as the tensile strength.

4.4.2.5 Water absorption. The test specimen shall be 2 inches square, and all surfaces and edges shall be smooth and free from burrs. The specimen shall be weighed (weight 1), then placed in distilled water at room temperature (75°F. plus or minus 5°F.) for 2 hours, removed, wiped free of surface moisture, and immediately reweighed (weight 2). The amount of water absorption shall be calculated as follows:

$$\frac{\text{Weight 2-Weight 1}}{\text{Weight 1}} \text{ X 100 percent water absorption}$$

4.4.2.6 Compression set.

4.4.2.6.1 Testing machine. Any standard type testing machine that is accurate to within 1 percent of the initial load may be used. The crosshead speed of the testing machine shall be such that the loads are accurately indicated, but shall not exceed 1 inch per minute. The disk holding fixture shall consist of a steel cylinder and a steel piston. The cylinder shall be 3.500 inches long by 2.500 inches outside diameter; ends shall be ground square with bore; bore shall be equal to the diameter of the disks listed in table II, plus 0.031 inch. The piston shall be 2.500 inches long with ends ground square with the axis and shall be a neat slide fit in the cylinder.

4.4.2.6.2 Test procedure. Unless otherwise specified (see 6.2), compression test specimens shall be cut in the form of circular disks of a size indicated in Table II. Select at random 22 disks, place in fixture in testing machine, put under an initial load of 100 pounds, set dial gage at zero, run load up to required amount, and then back to 100 pounds. The dial shall then indicate the amount of compression or set. One test only shall be made on each set of disks.

5. PACKING

5.1 Packaging & packing. Unless otherwise specified (see 6.2) packaging and packing shall be commercial in accordance with ASTM D 3951.

5.2 Marking. Unless otherwise specified (see 6.2) interior packages and shipping containers shall be marded in acordance with ASTM D 3951.

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6. NOTES

6.1 Intended use. Fiber material covered by this specification is intended for use in buffer assemblies of machine guns and rifles.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Dimensions, such as thickness, size, tolerance, and shape of material required (see 3.4).
c. If inspection requirements are different (see 4.1).
d. If different sample size for tests is required (see 4.2.3).
e. If different packaging and marking is required (see 5.1 and 5.2).

6.3 Metric units. When metric units are required, units for inch, pounds (force) and degree Fahrenheit may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by		Metric SI unit
inch	2.54	=	centimeter (cm)
pound (force)	4.448	=	newton (N)
degree Fahrenheit(^{o}F)	$t^{o}C=t^{o}F-32/1.8$	=	degree Celsius (^{o}C)

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - AL, AR, MI

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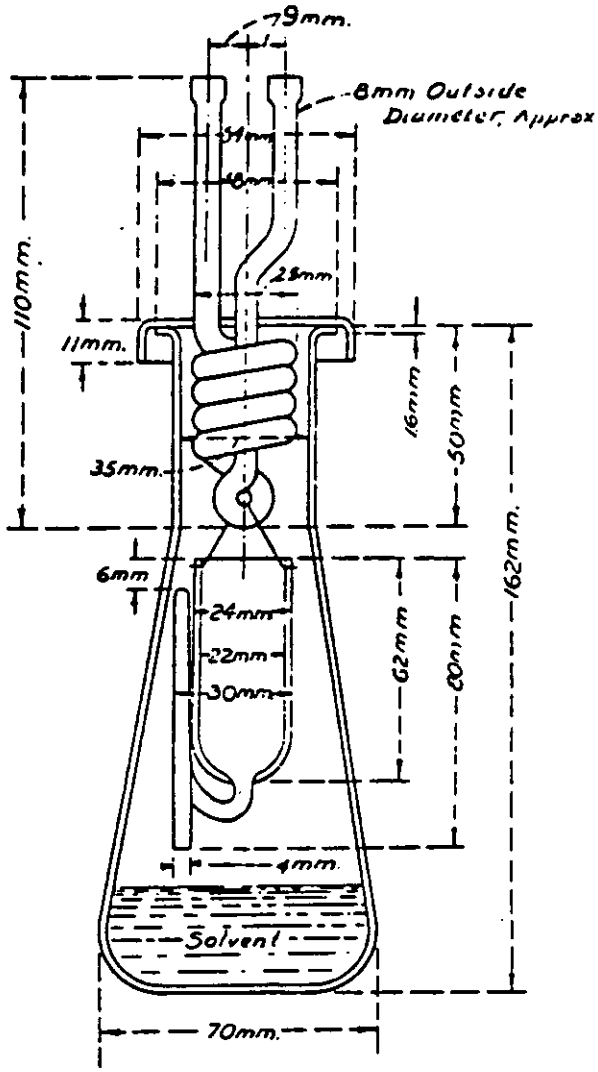


FIGURE 1 —Extraction apparatus.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-13526B	FIBER, RED (SHEET), FOR SMALL ARMS BUFFER DISKS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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